

PHASE 2 – REQUIREMENT ANALYSIS

PocketSmart AI: Your Smart Budget & Recommendation Assistant

1. Project Overview

PocketSmart AI is a web-based Generative AI application designed to help users plan budgets and receive personalized recommendations based on their requirements. The system can support scenarios such as home interior planning, event or party planning, and jewelry shopping. It combines planner features, adaptive forms, recommendation logic, and AI-assisted suggestions to make planning and decision-making easier.

2. Functional Requirements

Code	Requirement	Description
FR01	User Registration/Login	The system shall allow users to create an account and log in securely.
FR02	User Authentication	The system shall validate user credentials and maintain an authenticated session.
FR03	Data Storage	The system shall store required user, planner, recommendation, and history data.
FR04	Planner Functionality	The system shall allow users to enter requirements, budget, preferences, and planning details.
FR05	Recommendation Functionality	The system shall generate personalized recommendations based on user inputs.
FR06	Adaptive Forms	The system shall display relevant input fields according to the selected planning scenario.
FR07	Scenario Support	The system shall support planning scenarios such as home interior, events/parties, and jewelry shopping.
FR08	Recommendation Details	The system shall display useful details for recommended options.
FR09	History Viewing	The system shall allow users to view previously generated plans and recommendations.
FR10	Details Viewing	The system shall allow users to open and view saved plan/recommendation details.
FR11	Logout	The system shall allow users to log out and terminate the active session.

3. Non-Functional Requirements

Requirement	Description
Usability	The interface should be simple, clear, and easy for users to navigate.
Security	User authentication and application data should be handled securely. Secret keys and API credentials must not be exposed in source code.
Performance	The application should respond to normal user actions and recommendation requests within a reasonable time.

Requirement	Description
Reliability	The system should handle valid inputs consistently and preserve saved planner/history information.
Maintainability	The application should use a modular structure so that backend, database, frontend, and recommendation components can be updated independently.
Scalability	The design should allow additional planning scenarios and recommendation features to be added later.
Compatibility	The web application should work with commonly used modern web browsers.

4. Technology Requirements

- **Programming Language:** Python
- **Web Framework:** Flask / Python web framework
- **Frontend:** HTML, CSS and JavaScript
- **Database:** Database system for storing users, plans, recommendations and history
- **AI/Recommendation Layer:** Generative AI-assisted recommendation component
- **Dependency Management:** requirements.txt
- **Configuration:** env.example for environment variables and API configuration

5. User Requirements

- Users should be able to register and log in.
- Users should be able to select a planning category.
- Users should be able to enter their budget, preferences, and other required details.
- Users should receive recommendations based on the submitted requirements.
- Users should be able to view recommendation details.
- Users should be able to save and review previous plans and recommendations.
- Users should be able to log out securely.

6. System Requirements

Category	Requirement
Hardware	A computer/laptop capable of running Python and a modern web browser.
Software	Python, Flask, a code editor such as VS Code, database support, and a modern web browser.
Internet	Internet access may be required for AI/API-based recommendation services and external integrations.
Configuration	Required API keys, database settings, and secret values should be configured through environment variables.

7. Project Modules

Module	Purpose
Authentication Module	Registration, login, session management and logout.
Planner Module	Collects planning requirements, budget and preferences.
Recommendation Module	Processes user requirements and generates suitable recommendations.
Database Module	Stores and retrieves user, planner, recommendation and history information.
History Module	Displays previously created plans and recommendations.
Details Module	Shows detailed information for a selected plan or recommendation.
Frontend/UI Module	Provides web pages, forms, navigation and user interaction.

8. Expected Outcome

The completed system is expected to provide a simple web interface through which authenticated users can create planning requests, receive AI-assisted recommendations, view details, and access their previous planning history. The modular requirements defined above will guide the subsequent design, development, testing, and documentation phases.

Note: This Phase 2 document follows the requirement categories specified in the provided GitHub Repository upload format document, including functional, non-functional, and technology requirements.